

Completing the induction: why E/F is unramified in the $\mathbb{Z}/p^r\mathbb{Z}$ case

Working note

Abstract

This note gives a complete proof that “ $\mathcal{P}^{[\deg \mathfrak{n}]}$ is unramified at $\eta_{\mathfrak{n}}$ ” for $G = \mathbb{Z}/p^r\mathbb{Z}$ (the red-flagged gap). It first explains, via the field lattice and Proposition 23 (the canonical Cartesian square), how the hypotheses (H1),(H2) arise and *why they cannot finish* – the base change $K_{C'}/F$ is wildly ramified, so the inertia comparison is not onto (§§1–3). It then closes the proof by a different and unconditional route (§4): writing the symmetric-power cover as $\wp(\vec{Z}) = \bigoplus_{i=1}^d \vec{f}(x_i)$, the hypothesis “ramification $< d$ ” bounds the poles of the Witt datum, and – using only the isobaricity of Witt addition and an elementary symmetric-function estimate – forces *every* Witt component to be regular at η_C . Hence the cover is unramified there, for all r at once; the base case $r = 1$ is exactly the already-proven lemma. The single external input is the Artin–Schreier–Witt break formula.

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1 Notation and the field lattice

We use the notation of the “wrong path” section. Let $\mathcal{P} \rightarrow U = C \setminus \mathfrak{m}$ be a connected G -torsor, $G = \mathbb{Z}/p^r\mathbb{Z}$, with ramification at p_{∞} bounded by $d := \deg \mathfrak{n}$ (i.e. the d -th *upper-numbering* ramification group is trivial). After the standard reductions we may assume $k = \bar{k}$, that $\mathcal{P} \rightarrow U$ is *totally ramified* at p_{∞} , and that the inertia is all of G .

For the induction we use the decomposition

$$\mathcal{P} \longrightarrow \mathcal{P}' \longrightarrow U, \quad \mathcal{P} \rightarrow \mathcal{P}' \text{ is a } \mathbb{Z}/p\mathbb{Z}\text{-torsor,} \quad \mathcal{P}' \rightarrow U \text{ is a } \mathbb{Z}/p^{r-1}\mathbb{Z}\text{-torsor,}$$

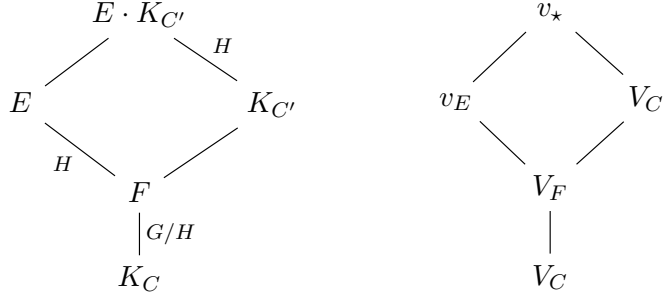
so that in the notation $H = \mathbb{Z}/p\mathbb{Z}$ (the unique order- p subgroup), $G/H = \mathbb{Z}/p^{r-1}\mathbb{Z}$, and $\mathcal{P}' = \mathcal{P}_{G/H}$. Taking function fields of the four corners of the canonical Cartesian square (Proposition 23) and writing $K_C = \text{Frac } V_C$, $K_{C'} = \text{Frac } V_{C'}$, we get

$$K(\mathcal{P}^{[d]_G}) = E, \quad K((\mathcal{P}_{G/H})^{[d]}) = K(\mathcal{P}'^{[d]}) = F, \quad K((\mathcal{P}_{G/H})^{(d)}) = K_{C'}, \quad K(U^{(d)}) = K_C,$$

with

$$E/F \text{ Galois of group } H = \mathbb{Z}/p\mathbb{Z}, \quad F/K_C \text{ Galois of group } G/H = \mathbb{Z}/p^{r-1}\mathbb{Z}.$$

Because $\mathcal{P}^{[d]H}$ is integral, the fourth corner has the single function field $E \cdot K_{C'} = E \otimes_F K_{C'}$, and $E, K_{C'}$ are linearly disjoint over F : $E \cap K_{C'} = F$. The relevant places are V_C (at η_C), $V_{C'}$ (at $\eta_{C'}$), $V_F := V_{C'}|_F$, a place v_E of E over V_F , and a place $v_\star$ of $E \cdot K_{C'}$ over both v_E and $V_{C'}$:



2 What Proposition 23 buys: the two hypotheses

(H1) from the inductive hypothesis. $\mathcal{P}' \rightarrow U$ is a $\mathbb{Z}/p^{r-1}\mathbb{Z}$ -torsor, still totally ramified at p_∞ with break $< d$ (upper numbering is compatible with the quotient $G \twoheadrightarrow G/H$). By the inductive hypothesis,

$$(H1) \quad \mathcal{P}'^{[d]} \text{ is unramified at } \eta_C \iff F/K_C \text{ is unramified at } V_C.$$

In particular $e(V_F | V_C) = 1$.

(H2) from Proposition 23 and the base case. The Cartesian square identifies $\mathcal{P}^{[d]H}$ with the fibre product, hence $E \cdot K_{C'} = E \otimes_F K_{C'}$, and exhibits $\mathcal{P}^{[d]H} \rightarrow (\mathcal{P}_{G/H})^{(d)}$ as the d -th symmetric power of the $\mathbb{Z}/p\mathbb{Z}$ -torsor $\mathcal{P} \rightarrow \mathcal{P}'$ over the curve C' . Its break at the point over p_∞ is still $< d$ (lower numbering restricts to subgroups: $H_i = G_i \cap H$, and $G_d = 1$). Applying the *already proven* $r = 1$ theorem with base curve C' :

$$(H2) \quad \mathcal{P}^{[d]H} \text{ is unramified at } \eta_{C'} \iff E \cdot K_{C'}/K_{C'} \text{ is unramified at } V_{C'}.$$

Goal. By (H1), the desired conclusion “ $\mathcal{P}^{[d]G}$ unramified at η_C ”, i.e. “ E/K_C unramified at V_C ”, is equivalent to

$$\boxed{E/F \text{ is unramified at } V_F}$$

a single $\mathbb{Z}/p\mathbb{Z}$ statement. This is the entire remaining content of the theorem.

3 Why (H2) does not finish: the obstruction, quantitatively

The two extensions E/F and $E \cdot K_{C'}/K_{C'}$ are related by restriction $\text{Gal}(E \cdot K_{C'}/K_{C'}) \xrightarrow{\sim} \text{Gal}(E/F) = H$, under which the inertia injects

$$I(v_\star | V_{C'}) \hookrightarrow I(v_E | V_F).$$

Hypothesis (H2) says the source is trivial; the goal says the target is trivial. The injection is onto only if the base change $K_{C'}/F$ is unramified at $V_{C'} | V_F$ – but it is not:

Lemma 1 (wild ramification of the base change). *At $V_{C'} \mid V_F$ one has $e(V_{C'} \mid V_F) = p^{r-1}$.*

Proof. C'/C is totally ramified of degree p^{r-1} at p_∞ , so with a local parameter w at the point p'_∞ and s the parameter at p_∞ we have $s = (\text{unit}) w^{p^{r-1}}$. With $\sigma_j = e_j(s_1, \dots, s_d)$ (coordinates at V_C) and $\tau_j = e_j(w_1, \dots, w_d)$ (coordinates at $V_{C'}$), the Frobenius identity in characteristic p gives

$$e_j(w_1^{p^{r-1}}, \dots, w_d^{p^{r-1}}) = e_j(w_1, \dots, w_d)^{p^{r-1}} = \tau_j^{p^{r-1}},$$

hence $\sigma_j = c^j \tau_j^{p^{r-1}} + (\text{higher } \tau\text{-order})$. Distinct monomials σ^α have distinct leading terms $\tau^{p^{r-1}\alpha}$, so for every $g \in K_C$, $V_{C'}(g) = p^{r-1} V_C(g)$, i.e. $e(V_{C'} \mid V_C) = p^{r-1}$. With $e(V_F \mid V_C) = 1$ from (H1), $e(V_{C'} \mid V_F) = p^{r-1}$. $\square$

Thus $K_{C'}/F$ is *wildly* ramified of degree p^{r-1} at $V_{C'} \mid V_F$. Such a base change can absorb the ramification of E/F (the wild analogue of Abhyankar): (H2) holding is consistent with E/F being ramified. Concretely, the place where the inertia injection fails to be onto is exactly the place where the symmetric-power base change q introduces its own wild ramification at infinity. **Conclusion:** (H1)+(H2)+Proposition 23 cannot prove the goal; the hypothesis “ramification $< d$ ”, so far used only to obtain (H1) and (H2), must be used a second time to bound E/F at V_F *directly*.

4 The completion: regularity of the top Witt component

Reduce “unramified” to “regular”. Since E/F is a $\mathbb{Z}/p\mathbb{Z}$ -extension in characteristic p , it is never tame, so

$$E/F \text{ unramified at } V_F \iff \text{its Artin-Schreier class } \beta \in F \text{ is regular at } V_F$$

(i.e. $\beta \equiv (\text{element of } \mathcal{O}_{V_F}) \bmod \wp(F)$, where $\wp(z) = z^p - z$). By the $r = 1$ proof, the local field at η_C is

$$\widehat{K}_{\eta_C} = \kappa(\eta)((e_d)), \quad \kappa(\eta) = k\left(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d}\right), \quad e_d = \prod_i x_i,$$

and the cover $\mathcal{P}^{[d]G}$ is the Artin-Schreier-Witt extension $\wp(\vec{Z}) = \vec{\alpha}$, where $\vec{\alpha} = \sum_{i=1}^d \vec{f}(x_i) \in W_r(\widehat{K}_{\eta_C})$ and $\vec{f} = (f_0, \dots, f_{r-1})$ is the Witt datum of $\mathcal{P}$ at p_∞ .

Rather than descend through the wild base change of §3, we bound the poles of *every* Witt component of $\vec{\alpha}$ directly. This bypasses the obstruction entirely and proves the theorem for all r at once.

Input: the Artin-Schreier-Witt break formula. We take $\vec{f} = (f_0, \dots, f_{r-1})$ to be the principal part at p_∞ (in a local parameter x with $p_\infty = \{x = 0\}$, residue field k perfect), and write $m_j := -v_x(f_j) \geq 0$ for the pole order of f_j . The only external input is the classical computation of the ramification of Artin-Schreier-Witt extensions.

Proposition 2 (Schmid-Witt break formula). *Let $K = k((x))$ with k perfect of characteristic p , and let L/K be the $\mathbb{Z}/p^r\mathbb{Z}$ -extension defined by $\vec{f} \in W_r(K)$ via Artin-Schreier-Witt theory. Then the largest upper-numbering ramification break of L/K equals*

$$u = \min_{\vec{f}} \max_{0 \leq j \leq r-1} p^{r-1-j} m_j,$$

the minimum taken over representatives of the class of $\vec{f}$ modulo $\wp W_r(K)$ and attained on a reduced representative. For $r = 1$ this is the break $= m$ of [2, Thm.] (the case used in the existing $r = 1$ proof).

This is the Schmid formula [1] in its Witt-vector form; the ramification filtration of W_r -extensions and this break are computed in [2], and the break formula in exactly the displayed shape (valid over any perfect residue field) is [3]. We use only the easy consequence: since the break is the *minimum* over representatives, the hypothesis “ramification bounded by d ”, i.e. $u < d$, lets us **fix a representative $\vec{f}$ with**

$$p^{r-1-j} m_j < d \quad \text{for all } 0 \leq j \leq r-1, \quad \text{equivalently} \quad m_j < d p^{j-(r-1)}. \quad (1)$$

Replacing $\vec{f}$ by a $\wp$ -equivalent representative changes $\vec{\alpha} = \bigoplus_i \vec{f}(x_i)$ only by $\wp$ of a Witt vector (since $\wp$ is additive and commutes with $\sum_i$), hence does not change the cover $\mathcal{P}^{[d]G}$; so we are free to use this $\vec{f}$.

Two elementary lemmas. Throughout put $y_i := 1/x_i$, so each $f_j(x_i)$ is a polynomial in y_i of degree m_j , and $\vec{\alpha}$ is a Witt vector of *polynomials* in $y_1, \dots, y_d$, symmetric under S_d .

Lemma 3 (symmetric regularity). *Let $\Phi \in k[y_1, \dots, y_d]^{S_d}$ be symmetric, with every monomial of total degree $< d$. Then $\Phi \in \kappa(\eta)$; in particular Φ is regular at η_C (i.e. $V_C(\Phi) \geq 0$).*

Proof. Φ is a k -combination of monomial symmetric functions $m_\lambda(y)$ with $|\lambda| < d$. Each m_λ is a polynomial in the elementary symmetric functions $e_\mu(y) = \prod_i e_{\mu_i}(y)$ with $\mu \vdash |\lambda|$; every part $\mu_i \leq |\lambda| < d$, so only $e_1(y), \dots, e_{d-1}(y)$ occur – never $e_d(y)$. Now $e_k(y) = e_{d-k}(x)/e_d(x)$, and for $1 \leq k \leq d-1$ this has $V_C(e_{d-k}(x)) = V_C(e_d(x)) = 1$, hence $e_k(y) \in \kappa(\eta) = k(e_1/e_d, \dots, e_{d-1}/e_d)$. Therefore $\Phi \in \kappa(\eta) \subset \hat{\mathcal{O}}_{\eta_C}$. $\square$

Lemma 4 (isobaricity of Witt addition). *Give the variable $f_j(x_i)$ the weight p^j . Then the component $\alpha_n = (\bigoplus_{i=1}^d \vec{f}(x_i))_n$ is isobaric of weight p^n : each of its monomials $\prod_{i,j} (f_j(x_i))^{e_{ij}}$ satisfies $\sum_{i,j} e_{ij} p^j = p^n$.*

Proof. The n -th ghost component $w_n(\vec{\alpha}) = \sum_{j=0}^n p^j a_j^{p^{n-j}}$ is isobaric of weight p^n for $\deg a_j = p^j$, and the universal Witt-addition polynomials S_n are determined by additivity of the ghosts; hence each S_n – and so each $\alpha_n = S_n(\{f_j(x_i)\})$ for the d -fold sum – is isobaric of weight p^n . $\square$

Theorem 5 (completion of the induction). *With $\vec{f}$ chosen so that (1) holds, every component α_n ($0 \leq n \leq r-1$) is regular at η_C . Consequently $\mathcal{P}^{[d]G}$ is unramified at η_C , proving the theorem for $G = \mathbb{Z}/p^r\mathbb{Z}$.*

Proof. Fix $n \leq r-1$ and a monomial $\prod_{i,j} (f_j(x_i))^{e_{ij}}$ of α_n ; set $E_j := \sum_i e_{ij}$, so by Lemma 4 $\sum_j E_j p^j = p^n$. Expanding each $f_j(x_i)$ (a polynomial in y_i of degree m_j), the monomial becomes a polynomial in the y_i all of whose terms have total degree

$$\leq \sum_{i,j} e_{ij} m_j = \sum_j E_j m_j \stackrel{(1)}{<} \sum_j E_j d p^{j-(r-1)} = d p^{-(r-1)} \sum_j E_j p^j = d p^{n-(r-1)} \leq d,$$

the last inequality because $n \leq r-1$. Thus *every* monomial of the symmetric polynomial α_n has total y -degree $< d$, so $\alpha_n \in \kappa(\eta)$ is regular by Lemma 3. Hence $\vec{\alpha} \in W_r(\hat{\mathcal{O}}_{\eta_C})$, and $\wp(\vec{Z}) = \vec{\alpha}$ is a finite étale (= unramified) $\mathbb{Z}/p^r\mathbb{Z}$ -cover of $\hat{\mathcal{O}}_{\eta_C}$ (the derivative of $\wp(Z) = Z^p - Z$ is the unit -1). Therefore $\mathcal{P}^{[d]G}$ is unramified at η_C . $\square$

Remark 6. The estimate is exact, not merely sufficient: because the weight $p^{j-(r-1)}$ is paired against the constraint $\sum_j E_j p^j = p^n$, the bound $d p^{n-(r-1)}$ is independent of how the carries distribute the exponents E_j . This is the precise form of the “break-scaling by $1/d$ ”: the bound $< d$ at p_∞ becomes < 1 at η_C on every Witt level simultaneously. Note also that Theorem 5 *subsumes* hypothesis (H1) (the components $\alpha_0, \dots, \alpha_{r-2}$, which cut out F/K_C , are regular) and never invokes the wild descent of §3: the field/Proposition-23 picture motivates the reduction, but the proof closes by bounding all components at once.

5 Summary

- (1) Proposition 23 reduces the $\mathbb{Z}/p^r\mathbb{Z}$ theorem to a single $\mathbb{Z}/p\mathbb{Z}$ statement, E/F unramified at V_F (§2), and exhibits the obstruction: the base change $K_{C'}/F$ is wildly ramified of degree p^{r-1} at $V_{C'} \mid V_F$ (Lemma 1), so (H1)+(H2) alone cannot finish (§3).
- (2) The decisive use of the unused hypothesis “ramification $< d$ ” is the pole bound (1). Combined with the isobaricity of Witt addition (Lemma 4) and symmetric regularity (Lemma 3), it forces *every* Witt component α_n of $\vec{\alpha} = \bigoplus_i \vec{f}(x_i)$ to be regular at η_C (Theorem 5), so $\mathcal{P}^{[d]_G}$ is unramified at η_C – for all r , the $r = 1$ case being the existing lemma.
- (3) The one external input is the Artin–Schreier–Witt break formula (Proposition 2); everything else is the elementary degree count above.

References

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- [2] L. Thomas, *Ramification groups in Artin–Schreier–Witt extensions*, J. Théor. Nombres Bordeaux **17** (2005), no. 2, 689–720.
- [3] G. G. Elder and K. Keating, *Artin–Schreier–Witt extensions and ramification breaks*, preprint (2025), arXiv:2503.16830.